Assignments

1. Introduction

Please read the document very carefully. If you have questions ask in the telegram group of the course (link can be found in Canvas).

You have to select 2 out of 4 tasks and complete them individually. You will need to select from the listed below tasks. You must select exactly 2 of 4 tasks.

1. **TextDetox**: Multilingual Text Detoxification
2. **KGQA**: Knowledge Graph Question Answering
3. **DimABSA**: Dimensional Aspect-Based Sentiment Analysis
4. **LoresMT**: MT for low-resource languages

Use this template to complete your assignment and upload it to Canvas: https://colab.research.google.com/drive/1ttPT6X4K0ovgbzmNjlcEiprkj1LaBuF2#scrollTo=8a46ab45-d215-41af-b910-63ff4a215a07 . The submitted file shall have the name corresponding to task name e.g., TextDetox.ipynb, KGQA.ipynb, etc. If file has a different name 1 point will be deducted from your final score.

You will have to submit assignment to the respective deadline. You need to submit the task by the deadline of 1st or 2nd assignment, respectively. To make it clear, you can submit in any order, e.g., 1st assignment deadline you can submit Task 3 and 2nd assignment deadline you can submit Task 2, but you can do another way around as well: for the 1st assignment you can submit Task 2 solution and for the 2nd task assignment you can submit Task 3 solution.

Assignments are supposed to be solvable in the standard Google Colab environment or a consumer-grade GPU like NVIDIA 3090 RTX. If for some reason this is not feasible let us know.

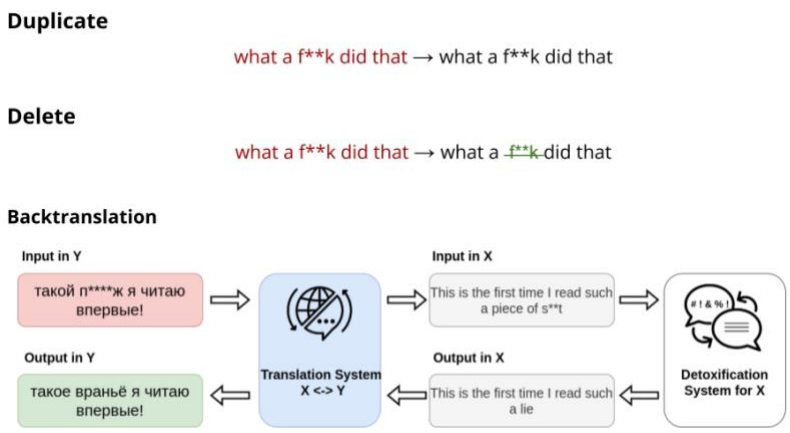
You are supposed to find the corresponding baselines for each task.

To solve the task, you are free to use any method, but naturally Transformer-based methods covered in the course are of priority as this is a task to practice such methods. At the same time, it may be useful to get rule-based elements. The use of proprietary LLMs is discouraged as all solutions must be reproducible unless you want to ship a solution with your private token with some balance on it.

2. Multilingual Text Detoxification

Task type: This is a multilingual sequence-to-sequence task.

Short description of the task: Given a toxic piece of text, re-write it in a non-toxic way while saving the main content as much as possible.



Evaluation: submission to leaderboard “Post-evaluation” phase. Indicate your team as “Skoltech--26”.

Full instructions including pointers to code, datasets, evaluation scripts**:**

https://pan.webis.de/clef25/pan25-web/text-detoxification.html

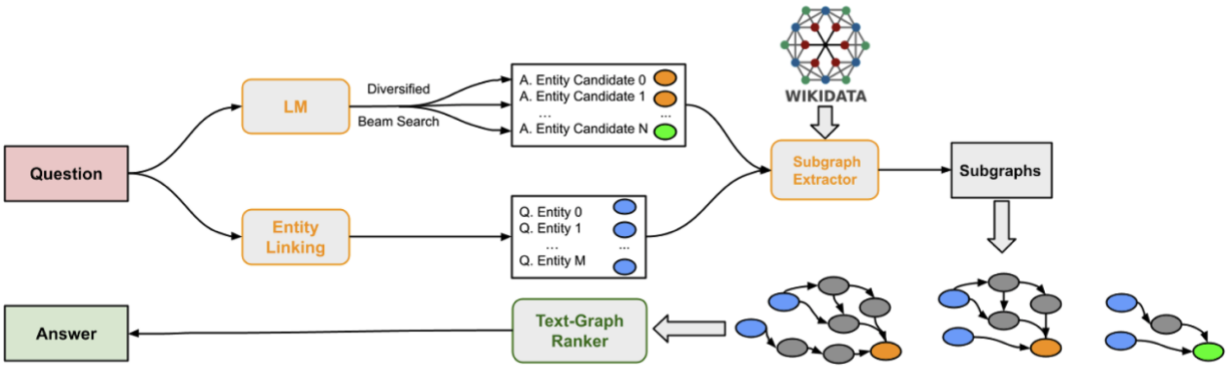
Papers with description of existing solutions, task formulation, and related work**:**

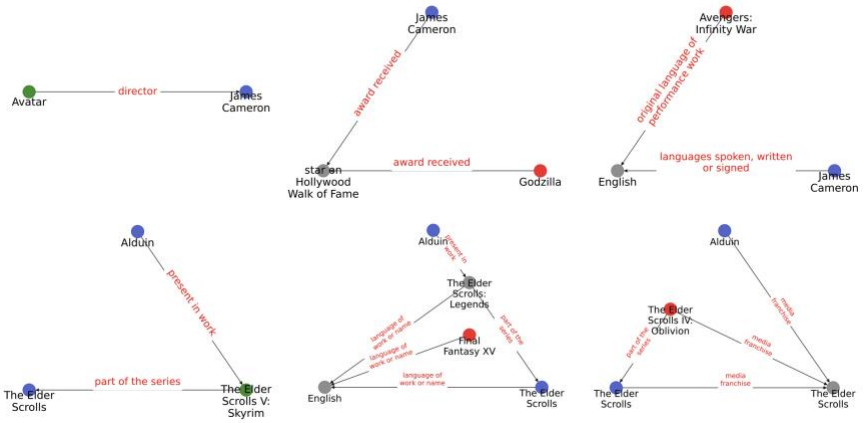
- https://ceur-ws.org/Vol-4038/paper_278.pdf
- https://ceur-ws.org/Vol-4038/ (search for papers containing “Detox” in the title).

3. Knowledge Graph Question Answering

Task type: This is a cross-modal (text+graph) categorization task.

Short description of the task: Given a pair of question-answer and knowledge graph representing a path say if the answer candidate is correct or not.





Evaluation: submission to leaderboard “Post-evaluation” phase. Indicate your team as “Skoltech--26”.

Full instructions including pointers to code, datasets, evaluation scripts**:**

https://sites.google.com/view/textgraphs2024/home/shared-task

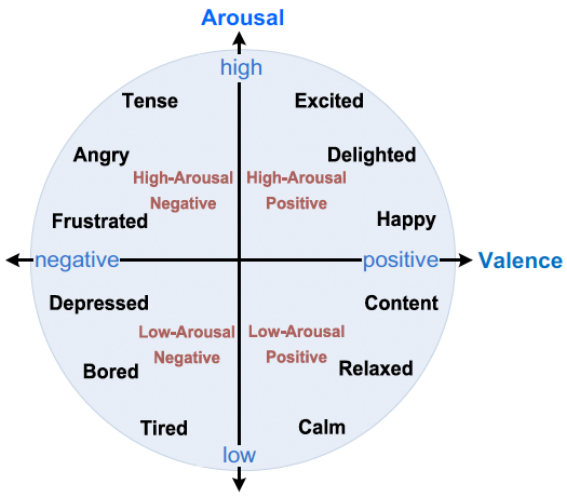
Papers with description of existing solutions, task formulation, and related work**:**

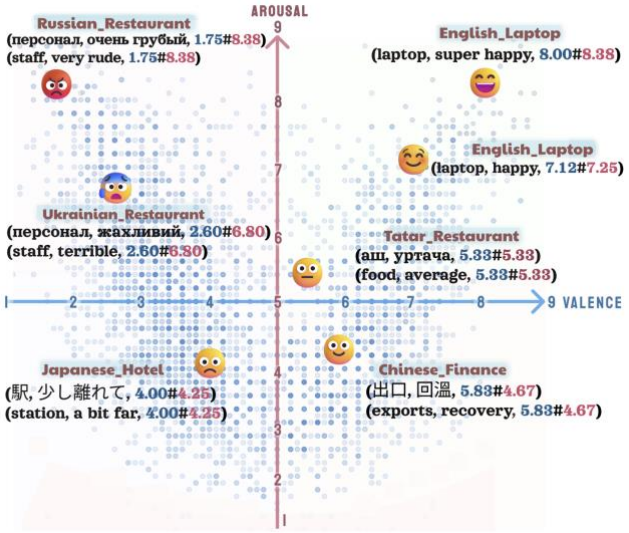
- https://aclanthology.org/2024.textgraphs-1.9/
- https://aclanthology.org/volumes/2024.textgraphs-1/ (search for papers containing “Shared Task” in the title).

4. Dimensional Aspect-Based Sentiment Analysis

Task type: This is a cross-modal (text+graph) categorization task.

Short description of the task: Perform detection of sentiment aspects in text like “movie, bad” indicating additionally values of arousal and valence associated with aspects.





Evaluation: submission to leaderboard is not available. Generate a table with results locally using evaluation scripts.

Full instructions including pointers to code, datasets, evaluation scripts**:**

https://github.com/DimABSA/DimABSA2026

Papers with description of existing solutions, task formulation, and related work**:**

- https://arxiv.org/abs/2601.23022

5. MT for low-resource languages

Task type: This is a multilingual sequence-to-sequence task.

Short description of the task: Perform machine translation for the following pairs of languages: Russian-Bashkir, English-Tatar, Russian-Kazakh, Russian-Kyrgyz.

Evaluation: submission to leaderboard is not available. Generate a table with results locally using evaluation scripts.

Full instructions including pointers to code, datasets, evaluation scripts**:**

https://ods.ai/tracks/turkic-lores-mt

Papers with description of existing solutions, task formulation, and related work**:**

- TBD